

COURSE TITLE : CALIBRATION & SIMULATION LAB
COURSE CODE : 5219
COURSE CATEGORY : P
PERIODS/WEEK : 4
PERIODS/SEMESTER: 52
CREDITS : 2

On completion of this course the student will be able

1. Familiarise HART communicator calibrator.
2. Connect HART communicator to any one of transmitter(DpTx,PTx,TTx) loop and perform basic set ups.
3. Set up a 4-20mA loop and find the value of the resistance for various percentages of process variable(using 24V dc source, mA meter & Potentiometer)
4. Calibrate SMART Temperature Transmitter for Different Ranges of Temperature using RTD sensor and perform.
 - A. Basic setup
 - B. Calibration
5. Calibrate SMART Temperature Transmitter for Different Ranges of Temperature using thermo couples and perform
 - A. Basic setup
 - B. Calibration
6. Perform control valve stroke test using 4-20 mA source .
7. Calibrate control Valve Positioner using 4- 20mA source.
8. Calibrate SMART Differential Pressure Transmitter for Different Ranges of Differential Pressure and perform
 - A. Basic setup
 - B. Calibration(Re range ,sensor trim, trim analog output)
9. Calibrate SMART Dp type level Transmitter for Different Ranges of DP and perform
 - A. Basic setup
 - B. Calibration(Re range, sensor trim, trim analog output
 - C. Elevation & suppression

10. Calibrate SMART Displacer type level Transmitter for Different Ranges of level and perform

A. Basic setup

B. Calibration (Re range, sensor trim, trim analog output)

C. Calibrate for Interface level measurement

11. Calibrate SMART Pressure Transmitter for Different Ranges Pressure and perform

A. Basic setup

B. Calibration (Re range, sensor trim, trim analog output)

USING MATLAB SOFTWARE

1. Simulation of half wave rectifiers.
2. Simulation of Full wave rectifiers (bridge & centre tap).
3. Simulation of half Controlled rectifiers .
4. Simulation of Full Controlled rectifiers (bridge & centre tap).
5. Plot Root locus.
6. Plot Bode plot
7. Simulation DC machines
8. Simulation of Inverter
9. Simulation of Choppers

VIRTUAL INSTRUMENTATION PRACTICES USING LAB VIEW

1. Add two numbers using numerical control.
2. Program to find the speed when the distance and time is given.
3. Program to calculate frequency(F) using time period(T).
4. Add two numbers and compute the sine of the result.
5. Suppose the temperature from a dial is in degree Celsius ,develop a program to convert this temperature to degree Fahrenheit .
6. Build a program to compare temperature with a set point and if it exceeds the limit, an indicator is ON.
7. Build a program to convert the temperature in degree C to degree K
8. Create a program that has a numeric control to input a number x, uses the add and multiply functions to calculate $3x^2 + 2x + 5$ and display the output using a numeric indicator.
9. Simulation of sinusoidal signal and display in the wave form graph. Change amplitude and frequency.
10. Simulate noise filled sinusoidal signal and filter it. Display both in one graph.